

REDBOX TECHNOLOGIES

INVOICE SOLUTION · FBR & PRA COMPLIANT POINT OF SALE, INVENTORY AND DIGITAL INVOICING

Ref. No.: _____

Date: _____

The Managing Director,
Tourism Development Corporation of Punjab (TDCP),
Lahore.

RESPONSE TO TECHNICAL QUERIES — POINT OF SALE & RESTAURANT MANAGEMENT SYSTEM, TDCP KALLAR KAHAR

Scope of the Proposed System

1. Company Profile, Relevant Experience and Contact Details

Company name:

Registration / NTN / STRN:

Registered office:

Focal person and designation:

Mobile / Email:

2. Presentation / Demo, Features and Technology Stack

n online

Modules

Layer	Coverage
Point of Sale	Dine-in, takeaway, delivery and room service orders; table and floor management; item modifiers; split and merge bills; held orders; authorised discounts; cash, card and mixed payment settlement
Kitchen	KOT printing to kitchen printers, Kitchen Display System, order status tracking
Rooms	Room categories, occupancy-linked billing, posting of restaurant charges to guest folio
Inventory	Multi-branch and multi-warehouse stock, stock movements with audit trail, purchase and GRN, supplier management, batch and expiry tracking, stock take and adjustment, low-stock alerts
Catalogue	Menu items, categories, pricing, barcode support, bulk import
Tax	FBR Digital Invoicing integration, PRA cloud POS integration, configurable tax rates, non-fiscal mode
Reporting	Daily sales, item and category-wise sales, cashier and shift reconciliation, payment-mode breakup, stock valuation, purchase and supplier ledgers, export to Excel and PDF
Administration	Multi-outlet hierarchy, role-based access control, complete audit log, user management
Customer display	Counter-side customer-facing order screen

Technology Stack

Component	Technology
Master database	PostgreSQL 16
Backend / API	Python (Django 5) with Django REST Framework — REST/JSON API
Admin web application	React (Vite, TypeScript, Tailwind CSS)
POS terminal	Electron with React, local SQLite for offline operation
Customer display	React
Background processing	Celery with Redis
Deployment	Linux server, Docker, Nginx, HTTPS/TLS

On Architectural Compatibility

The master database is PostgreSQL 16, the same database standardised by TDCP. All schema, tables and relational integrity reside in PostgreSQL, and the system can be hosted on TDCP's central PostgreSQL instance.

The front end is React, the same framework and developer skill set used in TDCP's existing systems.

The API layer is standard REST/JSON over HTTPS. Integration between systems takes place over HTTP and JSON, independent of the server-side runtime in which each endpoint is implemented. A Node or Express client consumes our API in the same manner as any other REST service.

The POS terminal is a desktop application with a local database. It continues taking orders and printing bills during internet or power interruption at the site, and reconciles automatically to the central database when connectivity is restored, with idempotent synchronisation so that no transaction is duplicated or lost.

The above is open to independent verification by TDCP's technical team or its development contractor.

3. Procurement Basis and Commercial Details

The solution is available on either basis:

Subscription: annual or monthly licence per outlet/terminal, inclusive of hosting, updates, FBR and PRA compliance maintenance and standard support.

One-time perpetual licence: one-time licence fee for deployment on TDCP infrastructure, with an optional Annual Maintenance and Support Contract.

A complete BOQ covering licence or subscription fee, implementation and configuration, data migration, on-site training, hardware where required (POS terminal, thermal and KOT printers, barcode scanner, cash drawer, customer display) and AMC will be furnished in TDCP's prescribed format.

There are no per-transaction charges and no third-party licence costs payable by TDCP beyond the BOQ.

4. Availability of APIs for Integration

Confirmed.

The system is built API-first. The web admin panel, POS terminal and customer display are all clients of the same REST API, and no function available in our own interface is unavailable over the API.

- Authenticated REST/JSON over HTTPS, with token-based authentication and role-scoped permissions
- Webhooks for event-driven push to TDCP systems, such as sale completion, stock threshold and day-end closure
- Integration with the E-Reservation portal to pull booking records for posting restaurant charges against the correct reservation, and to push consolidated charges back for settlement
- Direct database-level integration through read replicas, views or scheduled ETL into TDCP's central database
- OpenAPI/Swagger documentation provided

5. Proposed SLA, Support and Maintenance

Item	Commitment
Support hours	Business hours, with extended coverage during peak season
Critical incident (billing stopped)	Response within 1 hour, continuous effort until restoration
High (module unusable)	Response within 4 business hours
Medium and low	Response within 1 to 2 business days
Central system uptime	99.5% monthly, excluding notified scheduled maintenance
Outlet operation	Billing continues during internet or power interruption; outlet operation is not dependent on central uptime
Support channels	Telephone, WhatsApp, email and remote desktop assistance
Backup	Automated daily backup of the central database with defined retention and documented restoration procedure
Updates	Bug fixes and statutory FBR/PRA compliance updates at no additional cost during the subscription or AMC period
On-site attendance	Available for critical incidents not resolvable remotely
Escalation	Focal person, project lead, management — contact details furnished on award

Liquidated damages for SLA breach may be incorporated as per TDCP's standard contract terms.

6. Integration Requirements with TDCP's Existing Systems and Central Database

Our requirements from TDCP are:

1. API documentation of the E-Reservation and E-Payment portal, and of the existing RMS where applicable, with a technical point of contact
2. Access particulars for the central PostgreSQL instance, or agreement on a replication/ETL arrangement, along with TDCP's data governance and security requirements
3. A joint technical session with TDCP's development contractor to finalise endpoints, data ownership, master-data reconciliation, authentication and error handling
4. Network particulars at Kallar Kahar, including connectivity, LAN and firewall whitelisting requirements
5. Confirmation of the deployment model: hosted by us, hosted on TDCP infrastructure, or hybrid

The complete system can be deployed on TDCP's own infrastructure and central PostgreSQL database so that all data resides within TDCP's control.

7. Access to Database, Source Code and Technical Documentation

Database. Confirmed. TDCP retains full ownership of and access to all operational data. The system can be deployed on TDCP's own PostgreSQL server; where hosted by us, TDCP is provided direct database credentials or full scheduled replication, at TDCP's option. Complete data export in open formats is available at any time, including on termination.

Technical documentation. Confirmed. API documentation (OpenAPI/Swagger), database schema and data dictionary, system architecture documentation, deployment and configuration runbooks, backup and restoration procedures, and end-user manuals with training material for counter staff, kitchen staff and management.

Source code. The system is an existing licensed product rather than software written specifically for TDCP. We can proceed on any of the following, as determined by TDCP:

1. Source code escrow with a mutually agreed third-party agent, released to TDCP on defined trigger events including vendor default, insolvency or cessation of support
2. Source code delivered under a perpetual internal-use licence permitting TDCP to use, modify and maintain it for its own establishments
3. Full transfer of source code and intellectual property to TDCP

Options 2 and 3 carry a different commercial value from option 1, and separate costing will be submitted against whichever option TDCP selects.

We are available for a joint technical session with TDCP's IT team and development contractor, and for a live demonstration at your convenience.

Thank you for the opportunity to respond.

Yours faithfully,

Name: _____

Designation: _____

Redbox Technologies

Mobile / Email: _____